Sutanu Chatterjee

Academic CV • November 2026



Research Engineer, ITMO University
Doctoral researcher in Physical and Mathematical Sciences

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Research engineer in radiophysics and optoelectronics, combining RF and resonant wireless-power systems, photonics and sensing with scientific software and applied ML. I work from physical modelling and simulation through prototypes, measurement and real-use validation, alongside Python applications and PyTorch/TensorFlow models. My international collaboration spans Europe and Asia, including research visits to China since 2024.

Research and technical focus

APPLIED RESEARCH

- **Radiophysics**
  Resonant couplers
  RF characterisation
- **Optoelectronics**
  Perovskites and detectors
  Optical biosensing
- **Data and machine learning**
  Scientific data processing
  Design selection
- **Software and embedded**
  Engineering applications
  Audio signal processing

Research appointments

Research Engineer | ITMO University

2022–present • Department of Physics and Engineering • St. Petersburg

Wireless power transfer for low- and high-power applications, including coupler design, resonant metasurfaces, parameter estimation and control. Electromagnetic modelling in CST Studio Suite and Ansys, prototype construction, and validation with RF instrumentation.

Visiting Research Collaborator | Harbin Engineering University

Repeated visits since 2024 • Qingdao Innovation and Development Center • China

Collaborative research on wireless power transfer, metasurface resonators, and biomedical instrumentation. Co-author of published 2026 papers in IEEE T-MTT and IEEE Journal of Oceanic Engineering.

Research and Analysis | Ioffe Physical-Technical Institute

2020–2021 • St. Petersburg

Studied coastal-current effects on renewable energy generation in the Indian subcontinent.

Education and methods

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Education

PhD in Physical and Mathematical Sciences | ITMO University

2023–present • Ongoing

Optoelectronics and photonics for sensing applications. Advisor: Prof. Sergey Makarov. Russian Government Scholarship Quota.

MSc Electronics and Nanoelectronics | LETI

2019–2021 • Honours • GPA 4.92/5.00

Specialisation in photovoltaics and solar energy technology. Thesis on coastal-current effects on solar–wind generation and the economics of rural solar power in eastern India.

BTech Electronics and Communications Engineering | GNIT

2012–2016 • Kolkata, India

Among the top ten departmental scorers in the final semester. Thesis: BiRanger 1.0, a semi-autonomous robot with acoustic ranging, weather measurements, and remote control.

Methods

Data processing and machine learning. PyTorch, TensorFlow, MLPs, CNNs and surrogate models. pandas, NumPy, scikit-learn, SQL/PostgreSQL, OpenCV, PaddleOCR and Tesseract. Local-model AI agents and scientific data processing; hands-on AI/ML since 2022.

Simulation and design. CST Studio Suite, Ansys Suite, COMSOL, Autodesk Fusion 360 and Blender. Electromagnetic and multiphysics modelling; circuit simulation using LTspice; PCB design using KiCad and EasyEDA.

Programming and embedded systems. Python, C++/Embedded C, Ruby, JavaScript and Java; FastAPI, Flask and Django; Docker, Linux and Git/GitHub. pytest automation for serial COM and other protocols. STM32, ESP32, Arduino/AVR, Raspberry Pi, GNSS and LoRa.

Energy and geospatial analysis. HOMER Pro, PVsyst, HelioScope, QGIS, ArcGIS, NASA POWER and Solargis for energy-system modelling and environmental analysis.

Awards and training

- Russian Government Scholarship Quota for doctoral studies, ITMO University, 2023.
- Best International Graduate (Master's) Award, LETI, 2021.
- Master's with Honours (Red Diploma), GPA 4.92/5.00, 2021.
- Third-degree diploma in LETI's 2020/21 master's final-thesis awards.
- Merit-based full Russian Government Scholarship, 2019–2021.
- Zonal winner, Indo-US Robo-League 2014; qualified for the national level at IIT Bombay.

Robotics and Internet of Things, LETI (4 ECTS); AVR programming, Technophilia; Java Enterprise Edition, NVIT Kolkata (2016); Java Micro Edition, C Wizards Kolkata (2014).
Languages: English (native, C2), Bengali (native), Hindi (native), Russian (B2, working proficiency).

Applied software and engineering

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Selected client, research and embedded work, alongside public scientific tools. Client code and data remain private.



PROJECT 01

Client document automation

OpenCV preprocessing, PaddleOCR with Tesseract fallback, and Python/regex parsing into JSON for an accounts-payable team. Reduced manual data entry by approximately 40%.



PROJECT 02

RF surrogate modelling

PyTorch model for material thickness and RF reflection-loss targets. Traced simulation/prediction disagreement to a reversed material list, inspecting code and results with agentic AI tools.



PROJECT 03

Device testing and local AI

Python/pytest automation for serial COM and other protocols; bench and real-use embedded-device testing. Local-model AI agents connected to applications.

Public engineering applications

Sauryalic ATEARES; Matching Network Designer; WPT Unified Design Suite; EV Coil Parameter Estimator; Rectangular Planar Spiral Estimator.

ResonLab: acoustic resonator modelling and inverse design. PeroCast: model-based indoor photovoltaic and sensor-node energy budgeting.

Full software portfolio • GitHub

Journal articles and manuscripts

Published journal articles

J. Ye, **S. Chatterjee**, Z. Liu, B. Zhang, X. Liu, M. Song. **A Multimode Near-Field Radio Frequency Metasurface Resonator Transmitter for Multireceiver Power Transfer at 6.78 MHz**. *IEEE Transactions on Microwave Theory and Techniques*. Published online 9 September 2026. doi:10.1109/TMTT.2026.3729012

B. Zhang, C. Q. Jiang, Y. Lu, B. Cheng, F. Yang, L. Huang, W. Qin, **S. Chatterjee**, M. Song. **Underwater Energy Replenishment Platform with a Reconfigurable Magnetic Coupler for AUV Wireless Charging**. *IEEE Journal of Oceanic Engineering*. Published online 22 September 2026. doi:10.1109/JOE.2026.3722119

S. Chatterjee, A. Zolotarev, P. Terenteva, G. Baranov, P. Kapitanova. **Real-time estimator of parameters for coupled coils in an 11 kW wireless charging system for electric vehicles**. *Wireless Power Transfer*. 2025, vol. 12, article e027. doi:10.48130/wpt-0025-0019

A. V. Zolotarev, G. V. Demidova, A. S. Tsyrinova, P. V. Terenteva, G. D. Baranov, **S. Chatterjee**. **Synthesis of a simulation model of coupled coils in an electric vehicle wireless charging system**. *Russian Electrical Engineering*. 2024, vol. 95, no. 1, pp. 81–87. doi:10.3103/S1068371224010127

Manuscripts under review

These manuscripts are not yet accepted or published. Statuses updated October 2026.

Sutanu Chatterjee, Aleksandra Furasova, Sergey Makarov. **Machine-Learned Selection of the Size and Placement of Mie-Resonant Silicon Nanoparticles in Perovskite Solar Cells**. *Photonics and Nanostructures - Fundamentals and Applications*. Minor revisions requested; acceptance pending.

Sutanu Chatterjee, Vladislav Kalinichenko, Aleksandra Furasova, Sergey Makarov. **A Computational Framework for Perovskite Indoor Photovoltaic Energy Harvesting**. *Journal of Computational Electronics*. Under peer review.

Further publications

Book chapter

S. Chatterjee, Y. Zhao, Q. Zhou, D. Lukichev, A. Mamatov, and G. Demidova, Advanced power converters in wind energy systems: topologies, control strategies, and grid integration, in Advanced Wind Energy Systems: Grid Integration, Markets and Sustainability, 2026.

Conference papers

S. Chatterjee, X. Liu, and M. Song, Stochastic contact modeling of dry electrodes for common-mode rejection in brain-machine interfaces, 2026 IEEE International Applied Computational Electromagnetics Society Symposium (ACES-China), 2026. Accepted.

A. V. Zolotarev, E. A. Koreshin, S. Chatterjee, J. J. Justo, G. L. Demidova, P. V. Kapitanova, and R. C. Bansal, Performance validation of wireless power transfer for UAVs charging system considering misalignments, 2025 Fourth International Conference on Power, Control and Computing Technologies (ICPC2T), pp. 1 to 6, IEEE, 2025.

S. Chatterjee, A. Zolotarev, P. Terenteva, G. Baranov, and P. Kapitanova, Quick estimation of coupling coefficient in an electric vehicle wireless charging system, 2024 Antennas Design and Measurement International Conference (ADMInC), pp. 40 to 42, IEEE, 2024.

N. Kostik, A. Bobyl, R. Davydov, and S. Chatterjee, Modelling energy production and consumption in relation to climate parameters, 2021 International Conference on Electrical Engineering and Photonics (EExPolytech), pp. 273 to 277, IEEE, 2021. DOI 10.1109/EExPolytech53083.2021.9614785.

S. Barua and S. Chatterjee, Design of an automated and wirelessly controlled agricultural system for Siberian regions, Proceedings of the XIII Congress of Young Scientists, St. Petersburg, 8 to 11 April 2024.

Publications in Russian

А. В. Золотарев, Г. Л. Демидова, А. С. Цыринова, П. В. Терентьева, Г. Д. Баранов, С. Чаттерджи, Синтез имитационной модели связанных катушек в системе беспроводной зарядки электрического транспорта, Электротехника, № 1, с. 80 to 86, 2024.

А. В. Золотарев, Е. А. Корешин, А. С. Цыринова, С. Чаттерджи, П. В. Капитанова, Г. Д. Баранов, Система беспроводного питания датчиков, стационарно установленных в нефтегазовых скважинах, Наноиндустрия, т. 17, № S10-1 (128), с. 85 to 89, 2024.

А. В. Золотарев, Г. Д. Баранов, А. С. Цыринова, С. Чаттерджи, П. В. Терентьева, Г. Л. Демидова, П. В. Капитанова, Система беспроводной зарядки электротранспорта, Наноиндустрия, т. 17, № S10-1 (128), с. 90 to 95, 2024.

The Elektrotekhnika article is the Russian-language original of the 2024 Russian Electrical Engineering journal article.